

EE/CprE/SE 491 — sdmay27-37

Corn Harvest Data Collection Application

Week 01 Report

08/24/2026 – 09/22/2026

Client:	Bayer (Jason Knoche)
Faculty Advisor:	Dr. Phillip Jones

Team Members

Team Member	Role
Connor Moroney	Component Design Lead
Jonah Samuels	Client Interaction Lead
Carter Voelkel	Team Organization Lead
Megan Chng	Testing Lead
Cael Shroff	Documentation & Reporting Lead
Owen Gilbertson	Research Lead

Weekly Summary

This week was really about getting our footing. Specifically, we wanted to learn more about the project, each other, and our client. Since Jonah interns at Bayer, he already had a solid understanding of the project, so we leaned on him early to walk us through it, which helped a lot since the original project proposal was pretty vague on details. We also had our first full team meeting, where we did a quick team-building exercise and went around sharing our individual strengths and weaknesses, so we'd have a better sense of how to divide work going forward. The biggest priority, though, was connecting with our client to hear their side of the project and figure out what they actually need from us. That conversation helped clear up a lot of ambiguity and gave us a clearer picture of how our work will be used. By the end of the meeting, we'd nailed expectations, set up a recurring meeting schedule, and planned an in-person visit to Bayer's research lab to learn more about their work firsthand.

Past Week Accomplishments

Carter Voelkel

Created templates for future use like weekly_report_best_template. Researched that RFID's work for potential app integration to the backend. Researched SCiO pocket-sized NiR spectrometer for potential app integration via API & Bluetooth. The connectivity of these devices is important to reduce manual input of data by field data collectors when we make our app/website. Organization of the team's shared documentation for easier use.

Jonah Samuels

Arranged introductory meeting with Client. Scheduled future meetings with Client on a recurring basis. Reached out to advisor to schedule an introductory meeting. Brainstormed potential solutions to the engineering problem and potential implementations.

Connor Moroney

Gained access to the team website and GitHub repository. Researched how similar engineering problems have been solved in the past, looking for approaches that could be adapted or built on. Looked into several frameworks that could support the application, weighing their tradeoffs to help figure out which would be the best fit going forward.

Owen Gilbertson

Had introductory meeting with Jason, learned more about the “why” behind the project, and thought through some potential ideas for the project. (All still dependent on what the client wants or needs.)

Megan Chng

Brainstormed the specific features that the client wants, beginning the research on how to implement the features and any other technical parts of the application. Brainstormed data management solutions and analysis that need to be implemented.

Cael Shroff

Took notes during the introductory meeting with Jason, the client, and with the team. Noted the current process and what details and specifics are needed in the software solution. Helped brainstorm how to go about implementation and specific features for the project.

Group

Prepared discussion questions for our meeting with Jason Knoche to gain a deeper understanding of the project. Set up group communication and file-sharing channels, including text messaging and Microsoft Teams, to support ongoing collaboration. Participated in team-building exercises to better understand each team member's strengths and weaknesses. Discussed initial ideas and directions for the project going forward.

Pending Issues

Carter Voelkel

None reported.

Jonah Samuels

None reported.

Connor Moroney

One of the problems encountered was understanding what the input/output of the application would look like. The introductory meeting was helpful and gave a high-level understanding of the project, but more clarity is still needed on some of the specifics before real progress can be made.

Owen Gilbertson

Fully understanding the goal/application of what the team is to do. The meeting on Wednesday will definitely help clear some things up!!

Megan Chng

Currently still in the process of gathering high-level information; the full structure and architecture have yet to be decided and committed. More specifics and something a little more tangible are needed. The team also needs to look at the client's data worksheet to get an idea of exactly what needs to be built and accommodated for.

Cael Shroff

Still need to document and identify the scope of the project. There are a lot of things the team COULD do and add to the application as far as features go, but the team has yet to identify which items are more important and what can realistically be accomplished with the available time.

Individual Contributions

Name	Individual Contributions	Hours This Week	Cumulative Hours
Connor Moroney	Researching Frameworks/Existing Solutions	6	6
Jonah Samuels	Schedule Client/Advisor meetings, research	6	6
Carter Voelkel	Team Documentation/Organization, and research	6	6
Megan Chng	Research, Brainstorming backend solutions	5	5
Cael Shroff	Meeting Documentation, Feature brainstorming	6	6
Owen Gilbertson	Brainstorming future solutions discussed with Jason	6	6

Comments and Extended Discussion

No additional comments at this time.

Plans for the Upcoming Week

This week, the priority is finalizing the plan for the project and starting work on the lightning talk presentation. On the product side, we need to complete mock-ups for what the application will look like, and we need to make a decision on whether we're building this as a web app or a mobile app. That decision is foundational. It determines which frameworks and tools make sense for us going forward, and we'll be basing it on the research Connor has done this week along with the specific requirements the team gathers. The following is an outline of each team member's primary focus for the upcoming week:

Carter Voelkel

Work on lightning talk presentation.

Jonah Samuels

Confirm a time for advisor meeting; lightning talk presentation.

Connor Moroney

Get specific app requirements and start prototyping.

Owen Gilbertson

Get specific app requirements and start prototyping.

Megan Chng

Create a layout and start building the database.

Cael Shroff

Document team progress and first meeting with faculty advisor.

Summary of Weekly Advisor Meeting

We were unable to contact Dr. Jones. Therefore, we were unable to have an advisor meeting. We sent a follow-up email in hopes of scheduling a meeting for next week.